

Application of Information & Communication Technologies

**Introduction to Artificial
Intelligence (AI)**

Recap of Lecture 25

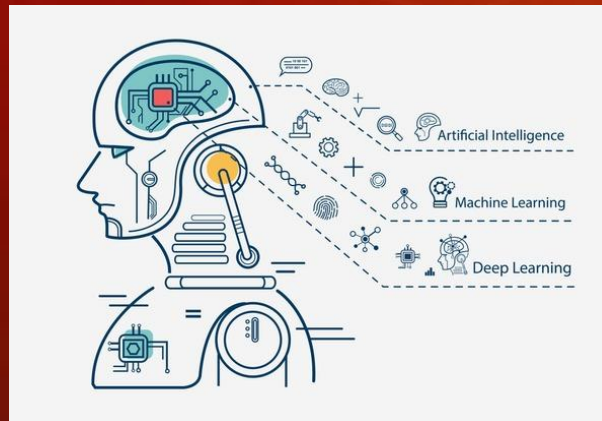
- ◆ Understand the importance of network security.
- ◆ Identify common cyber threats.
- ◆ Learn safe Internet and communication practices.
- ◆ Recognize ethical responsibilities in online communication.

Overview of Lecture 26

- ◆ Understand what Artificial Intelligence (AI) means
- ◆ Identify real-life uses of AI
- ◆ Differentiate between Narrow AI and General AI
- ◆ Discuss the benefits and challenges of AI
- ◆ Explain what Generative AI means
- ◆ Identify popular Generative AI tools
- ◆ Understand how to write good prompts
- ◆ Recognize ethical and safe use of AI

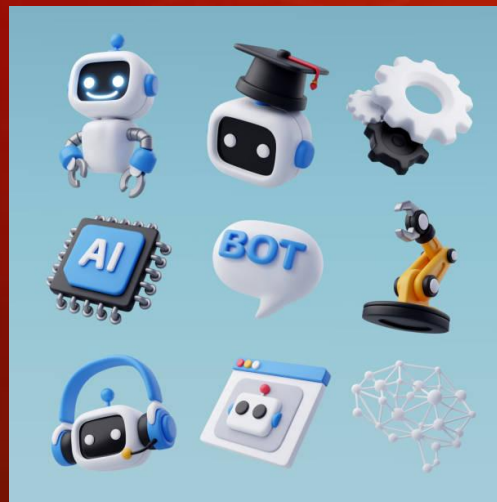
What is Artificial Intelligence (AI)?

- AI = Ability of machines to think, learn, and act like humans
- Uses data to solve problems and make decisions
- Examples → Siri answers your questions, Google suggests routes



AI in ICT Systems

- ◆ **AI Applications in ICT include:**
 - **Voice Assistants: Siri, Alexa, Google Assistant**
 - **Smart Devices: Smart TVs, Smart lights**
 - **Chatbots: Banks and websites answering customers**
 - **Face Recognition: Phone unlock or security systems**
 - **Email Filtering: Spam detection**



Examples of AI in Daily Life

Area

Navigation

Shopping

Entertainment

Healthcare

Education

Example of AI Use

Google Maps traffic updates

Amazon product recommendations

Netflix / YouTube suggestions

Disease detection / robotic surgery

Smart learning apps and AI tutors

Types of AI

Type	Description	Example
Narrow AI (Weak AI)	Works for one specific task	Siri, Google Search
General AI (Strong AI)	Thinks and acts like a human	Research robots / future AI

Advantages and Disadvantages of AI

Advantages

- Works 24/7 without tiring
- Reduces human error
- Fast decision making using big data
- Useful in medicine, transport, and finance

Disadvantages/ Challenges

- ◆ Expensive to build and maintain
- ◆ May replace human jobs
- ◆ Needs huge amounts of data
- ◆ Privacy and security concerns

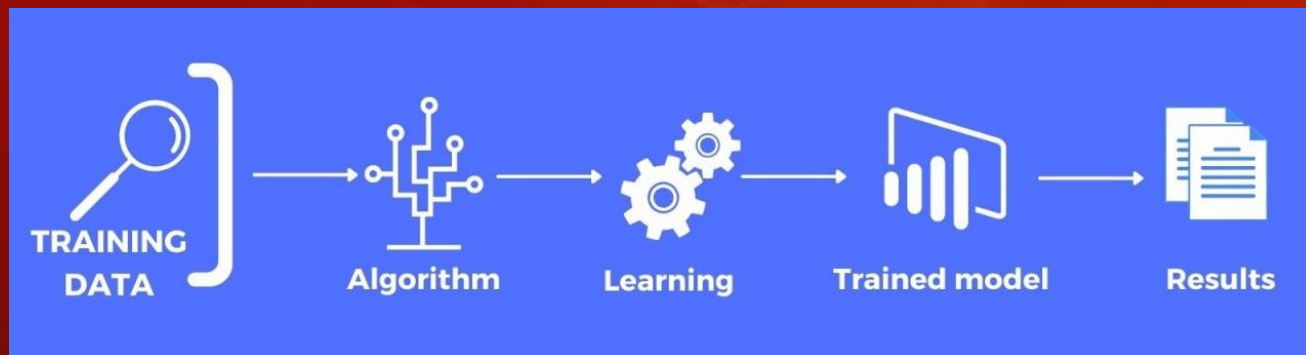
What is Generative AI?

- ◆ Generative AI = AI that creates new content
- ◆ Uses machine learning models to generate text, images, music, or video
- ◆ Works by learning patterns from large datasets



How Generative AI Works

- ◆ **Data Collection** – AI learns from millions of examples
- ◆ **Training** – Finds patterns and relationships
- ◆ **Generation** – Produces new text, image, or code based on input prompt



Examples of Generative AI Tools

Tool Name	Type	Output Example
ChatGPT	Text generation	Essays, explanations, code
DALL·E / Leonardo AI	Image generation	Drawings, digital art
Synthesia	Video generation	AI avatars & lectures
Soundful	Audio generation	AI music
Runway ML	Video & animation	Editing, creative tools

What is a Prompt?

- ◆ A prompt is a text instruction you give to an AI model
- ◆ The quality of your prompt determines the quality of the result
- ◆  **Example:**
 - **Prompt:** “Write a 50-word paragraph about climate change in simple English.”
 - **Output:** (AI-generated paragraph)



How to Write Good Prompts

- ◆ Be clear and specific
- ◆ Give context (what, who, how)
- ◆ Ask for format or style (paragraph, list, code, etc.)
- ◆ Test and refine your prompt
- ◆ Example Comparison:
 - Bad Prompt → “Explain AI”
 - Better Prompt → “Explain AI to a 12-year-old student with one example.”

Ethical and Responsible Use

- Always verify information from AI tools
- Avoid plagiarism and misinformation
- Respect privacy and intellectual property
- Use AI as an assistant, not a replacement for thinking



Advantages and Limitation of Generative AI

Advantages

- ◆ Saves time and boosts creativity
- ◆ Helps with learning, design, and content creation
- ◆ Available 24/7 for assistance

Limitations and Risks

- ◆ May generate incorrect or biased information
- ◆ Can be misused for fake news or deepfakes
- ◆ Depends on data quality and ethical use

Summary

- ◆ AI is changing how we live and work
- ◆ Used every day in ICT systems and devices
- ◆  Question for students:
 - “Where do you see AI in your daily life?”
- ◆ Generative AI creates new digital content
- ◆ ChatGPT and image generators are common examples
- ◆ Prompt engineering improves results
- ◆ Ethical use is key for safe AI practice